UCHIDA, Takaya

A Physical Oceanographer with a flair of data science

Office Address

Climate Dynamics Laboratory
Moscow Institute of Physics and Technology
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Languages

- Native: Japanese, English
- Conversational: French
- Learning: Russian

Education

Doctor of Philosophy, Physical Oceanography — October 2019
Columbia University in the City of New York, USA

Master of Philosophy, Physical Oceanography — February 2018
Columbia University in the City of New York, USA

Master of Arts, Physical Oceanography — May 2016
Columbia University in the City of New York, USA

Bachelor of Engineering, Ocean Engineering — March 2014
The University of Tokyo (東大), 日本

Research Experience

Senior Researcher, Physical Oceanography — March 2025 - present
Climate Dynamics Laboratory, МФТИ, Россия

Assistant Research Scientist, Physical Oceanography — December 2024
Center for Ocean-Atmospheric Prediction Studies, FSU, USA

MOPGA Postdoctoral Fellow, Physical Oceanography — September 2022
Institut des Géosciences de l'Environnement, CNRS, France

Publications

[M]: Work led by mentee

- Sun, L.[M], **T. Uchida**, T. Penduff, W. Dewar, B. Deremble, A. Poje, E. Chassignet & N. Wienders. (Submitted). On the dynamics of the subtropical mode water from an ensemble view. *J. Phys. Oceanogr.*, doi:10.22541/essoar.174802928.80389595/v1;

- **Uchida, T.**, A. Bodner, B. Reichl, A. Adcroft, B. Fox-Kemper, M. Ilicak, M. Bentsen, G. Marques & W. Large. (Submitted). Representation of surface mixed-layer eddies affects the large-scale ventilation of the global ocean. *Geophys. Res. Lett.*, doi:10.22541/essoar.174802994.41704767/v1;

- Yadidya, B., B. Arbic, J. Shriver, E. Zaron, M. Buijsman, L. Carrere, T. Lionel & **T. Uchida**. (Submitted). Predictability of phase-locked and non-phase-locked internal tides in the SWOT Cal/Val orbit. *Earth and Space Sci.*, doi:10.22541/essoar.174164232.20506498/v1;

- **Uchida, T.**, B. Yadidya, K. Lapo, X. Xu, J. Early, B. Arbic, D. Menemenlis, L. Hiron, E. Chassignet, J. Shriver & M. Buijsman. (Submitted). Dynamic mode decomposition of

geostrophically balanced motions from SWOT Cal/Val in the separated Gulf Stream. *Earth and Space Sci.*, doi:10.48550/arXiv.2407.09309;

- **Uchida, T.**, Q. Jamet, A. Poje, N. Wienders, L. Sun & W. Dewar. (2025). Dynamics and thermodynamics of the Boussinesq North Atlantic eddy kinetic energy spectral budget. *J. Adv. Model. Earth Sys.*, doi:10.1029/2024MS004781;

- **Uchida, T.**, Q. Jamet, W. Dewar, B. Deremble, A. Poje & L. Sun. (2024). Imprint of chaos on the ocean energy cycle from an eddying North Atlantic ensemble. *J. Phys. Oceanogr.*, doi:10.1175/JPO-D-23-0176.1;

- **Uchida, T.**, Q. Jamet, A. Poje, N. Wienders & W. Dewar. (2024). Wavelet-based wavenumber spectral estimate of eddy kinetic energy: Application to the North Atlantic. *Ocean Model.*, doi:10.1016/j.ocemod.2024.102392;

- **Uchida, T.**, Q. Jamet, A. Poje, N. Wienders, W. Dewar & B. Deremble. (2023). Wavelet-based wavenumber spectral estimate of eddy kinetic energy: Idealized quasi-geostrophic flow. *J. Adv. Model. Earth Syst.*, doi:10.1029/2022MS003399;

- Deremble, B., **T. Uchida**, W. Dewar & R. Samelson. (2023). Eddy-mean flow interaction with the Multiple Scale Quasi-Geostrophic Model. *J. Adv. Model. Earth Syst.*, doi:10.1029/2022MS003572;

- Chassignet, E., X. Xu, A. Bozec & **T. Uchida**. (2023). Impact of the New England seamount chain on the Gulf Stream pathway and variability. *J. Phys. Oceanogr.*, doi:10.1175/JPO-D-23-0008.1;

- **Uchida, T.**, D. Balwada, Q. Jamet, W. Dewar, B. Deremble, T. Penduff & J. Le Sommer. (2023). Cautionary tales from the mesoscale eddy transport tensor. *Ocean Model.*, doi:10.1016/j.ocemod.2023.102172;

- **Uchida, T.**, J. Le Sommer, C. Stern, R. Abernathey, C. Holdgraf, A. Albert, L. Brodeau, E. Chassignet, X. Xu, J. Gula, G. Roullet, N. Koldunov, S. Danilov, Q. Wang, D. Menemenlis, C. Bricaud, B. Arbic, J. Shriver, F. Qiao, B. Xiao, A. Biastoch, R. Schubert, B. Fox-Kemper, W. Dewar & A. Wallcraft. (2022). Cloud-based framework for inter-comparing submesoscale permitting realistic ocean models. *Geosci. Model Dev.*, doi:10.5194/gmd-15-5829-2022;

- **Uchida, T.**, Q. Jamet, W. Dewar, J. Le Sommer, T. Penduff & D. Balwada. (2022). Diagnosing the thickness-weighted averaged eddy-mean flow interaction from an eddying North Atlantic ensemble: The Eliassen-Palm flux. *J. Adv. Model. Earth Syst.*, doi:10.1029/2021MS002866;

- **Uchida, T.**, B. Deremble & S. Popinet. (2022). Deterministic model of the eddy dynamics for a midlatitude ocean model. *J. Phys. Oceanogr.*, doi:10.1175/JPO-D-21-0217.1;

- **Uchida, T.**, Q. Jamet, A. Poje & W. Dewar. (2021). An ensemble-based eddy and spectral analysis, with application to the Gulf Stream. *J. Adv. Model. Earth Syst.*, doi:10.1029/2021MS002692;

- Khatri, H., S. Griffies, **T. Uchida**, H. Wang & D. Menemenlis. (2021). Role of mixed-layer instabilities in the seasonal evolution of eddy kinetic energy spectra in a global submesoscale permitting simulation. *Geophys. Res. Lett.*, doi:10.1029/2021GL094777;

- **Uchida, T.**, B. Deremble & T. Penduff. (2021). The seasonal variability of the ocean energy cycle from a quasi-geostrophic double gyre ensemble. *Fluids*, doi:10.3390/fluids6060206;

- **Uchida, T.**, B. Deremble, W. Dewar & T. Penduff. (2021). Diagnosing the Eliassen-Palm flux from a quasi-geostrophic double gyre ensemble. In proceedings of the *EarthCube Annual Meeting*, doi:10.5281/zenodo.5496375. https://earthcube2021.github.io/ec21_book/notebooks/ec21_uchida_etal/notebooks/TU_05_Diagnosing-the-Eliassen-Palm-flux-from-a-quasi-geostrophic-double-gyre-ensemble.html;

- Jamet, Q., B. Deremble, N. Wienders, **T. Uchida** & W. Dewar. (2021). On wind-driven energetics of subtropical gyres. *J. Adv. Model. Earth Syst.*, doi:10.1029/2020MS002329;

- Jones, S., J. Busecke, **T. Uchida** & R. Abernathey. (2020). Vertical regridding and remapping of CMIP6 ocean data in the cloud. In proceedings of the *EarthCube Annual Meeting*, https://github.com/earthcube2020/ec20_jones_etal;

- **Uchida, T.**, D. Balwada, R. Abernathey, G. McKinley, S. Smith & M. Lévy. (2020). The impact of seasonality in eddy iron fluxes on primary production in the Southern Ocean. *Nature Comm.*, doi:10.1038/s41467-020-14955-0;

- **Uchida, T.**, D. Balwada, R. Abernathey, G. McKinley, S. Smith & M. Lévy. (2019). The contribution of submesoscale over mesoscale eddy iron transport in the open Southern Ocean. *J. Adv. Model. Earth Syst.,* doi:10.1029/2019MS001805;

- **Uchida, T.**, D. Balwada, R. Abernathey, C. Prend, E. Boss & S. Gille. (2019). Southern Ocean Phytoplankton Blooms Observed by Biogeochemical Floats. *J. Geophys. Res.: Oceans*, doi:10.1029/2019JC015355;

- **Uchida, T.**, R. Abernathey & S. Smith. (2017). Seasonality of eddy kinetic energy in an eddy permitting global climate model. *Ocean Model.,* doi:10.1016/j.ocemod.2017.08.006.

Non-refereed

- **Uchida, T**. (2019). *Seasonality in surface (sub)mesoscale turbulence and its impact on iron transport and primary production*. (Ph.D. dissertation, Columbia University in the City of New York). doi:10.7916/d8-9s8r-m049.

Mentoring

- Postdoctoral fellows: Luolin Sun (CNRS, France/FSU, USA); Habib Micaël Aguedjou (CNES, France);

- Doctoral students: Черносов Вячеслав Денисович (МФТИ, Россия); Ilkyeong Ma (FSU, USA);

Honors and Awards

- Co-investigator of the National Aeronautics and Space Administration (NASA) award '*Mesoscale eddies, internal waves, solitons in SWOT, global models, and downscaled regional models*' with the reference 80NSSC24K1649, funded for the duration of 2025-2028;

- Co-investigator of the National Science Foundation (NSF) award '*Novel Ensemble Based North Atlantic Diagnostics*' with the reference OCE-2123632, funded for the duration of 2022-2026;

- Co-investigator of the the French Les Enveloppes Fluides et l'Environnement (LEFE) award '*Assessing the Role of forced and internal Variability for the Ocean and climate Response in a changing climate (ARVOR)*' funded for the duration of 2022-2024;

- Attendee of Les Houches Summer School on Fundamental Aspects of Turbulent Flows in Climate Dynamics, which took place during August 2017 at l'École de Physique des Houches, France;

- Scholarship from the Heiwa Nakajima Foundation from September 2014 - August 2016.

- Laureate of the 2014 Dean's Award from the Faculty of Engineering, The University of Tokyo.

Technical Skills

- Proficient in compiling and running numerical models in Fortran, with experience using the MITgcm. Also experienced in tuning and running the Darwin biogeochemical model;

- Expert in big-data analysis and has continued contributing to the development of Python open-source software such as:

    1. `xrft`: doi:10.5281/zenodo.1402635 (https://xrft.readthedocs.io/en/latest/);
    2. `xwavelet`: doi:10.5281/zenodo.6984380 (https://github.com/roxyboy/xwavelet);
    3. `xgcm`: doi:10.5281/zenodo.4821276 (https://xgcm.readthedocs.io/en/latest/);

  available via the Github platform, and `pip` and `conda-forge` package managers. Also has contributed to the implementation of Pangeo Forge for cloud-based computational analyses (doi:10.5281/zenodo.6762536);

Service Activity

- Topic editor for the special edition '*In Memory of William Kurt Dewar: Exploring the Dynamics of Oceanic Boundary Currents and their Impact on Weather*' under Frontiers of Marine Science;

- Panelist for the NASA Research Opportunities in Space and Earth Science (ROSES) to review scientific proposals;

- Special U.S. government employee of NSF as a panelist and reviewer for reviewing scientific proposals;

- Reviewer for the Wallenberg Academy Fellowship awarded by the Knut and Alice Wallenberg Foundation in Sweden;

- Reviewer for the BIENVENÜE Call 2021 (H2020-MSCA-COFUND-2019) managed by the Regional Council of Brittany, France;

- Reviewed manuscripts for *J. Phys. Oceanogr.*, *J. Adv. Model. Earth Syst.*, *J. Geophys. Res.: Oceans*, *Geophys. Res. Lett.*, *Glob. Biogeochem. Cyc.*, *Nature Comm.*, *Ocean Model.*, *Climate Dyn., Fund. Res.*, *J. Marine Syst.*, *Biogeosci.*, *Geosci. Model Dev.*, *Front. Marine Sci.*, *Deep-Sea Res.: Part I*, *J. Limn. Oceanogr.*, *Fluids*, *J. Oper. Oceanogr.*, & *Inter. J. Digi. Earth*;

- Technician for the Lowered Acoustic Doppler Current Profiler (LADCP) measurements on the 2016 GO-SHIP 109N transect (https://currents.soest.hawaii.edu/go-ship/ladcp_rst_2015-2018/2016_I8S9N.html).